

Trapping Light in Diamond: The NV Center as a Room-Temperature Qubit

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DFT confirms NV⁻ as the thermodynamically stable, spin-triplet charge state in diamond, with a computed ZPL consistent with optical spin-readout for quantum sensing.

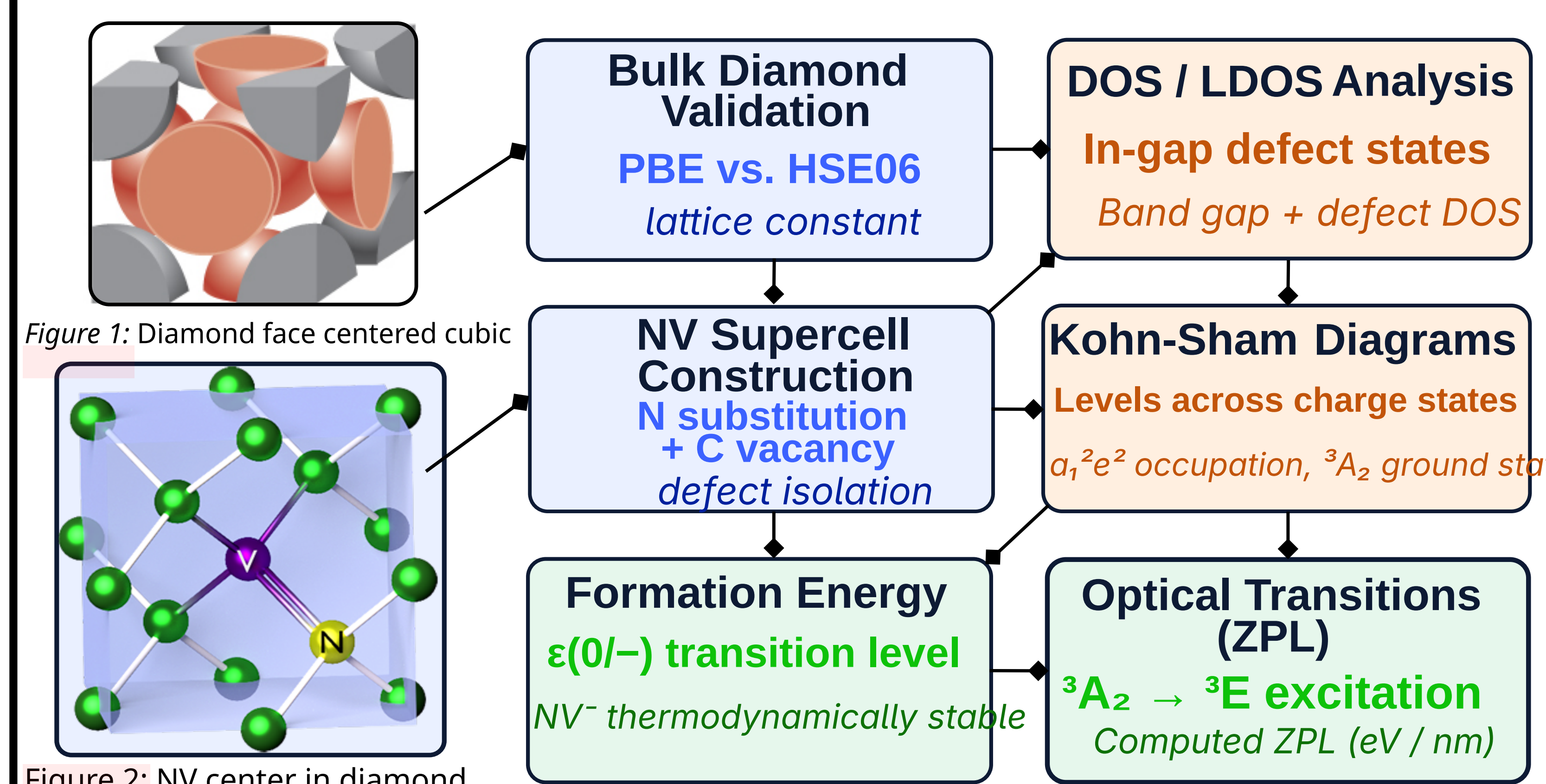
Why Diamond?

- Superconducting qubits need near-zero cooling — costly, hard to scale
- Point defects in wide-bandgap crystals offer a room-temperature alternative
- NV center forms when, inside diamond's carbon lattice, nitrogen replaces one carbon atom and a carbon is removed, leaving an vacancy.

NV center is a spin qubit that can act as both a qubit for computation and a nanoscale sensor, this project focuses on the properties enabling the optical spin-readout.

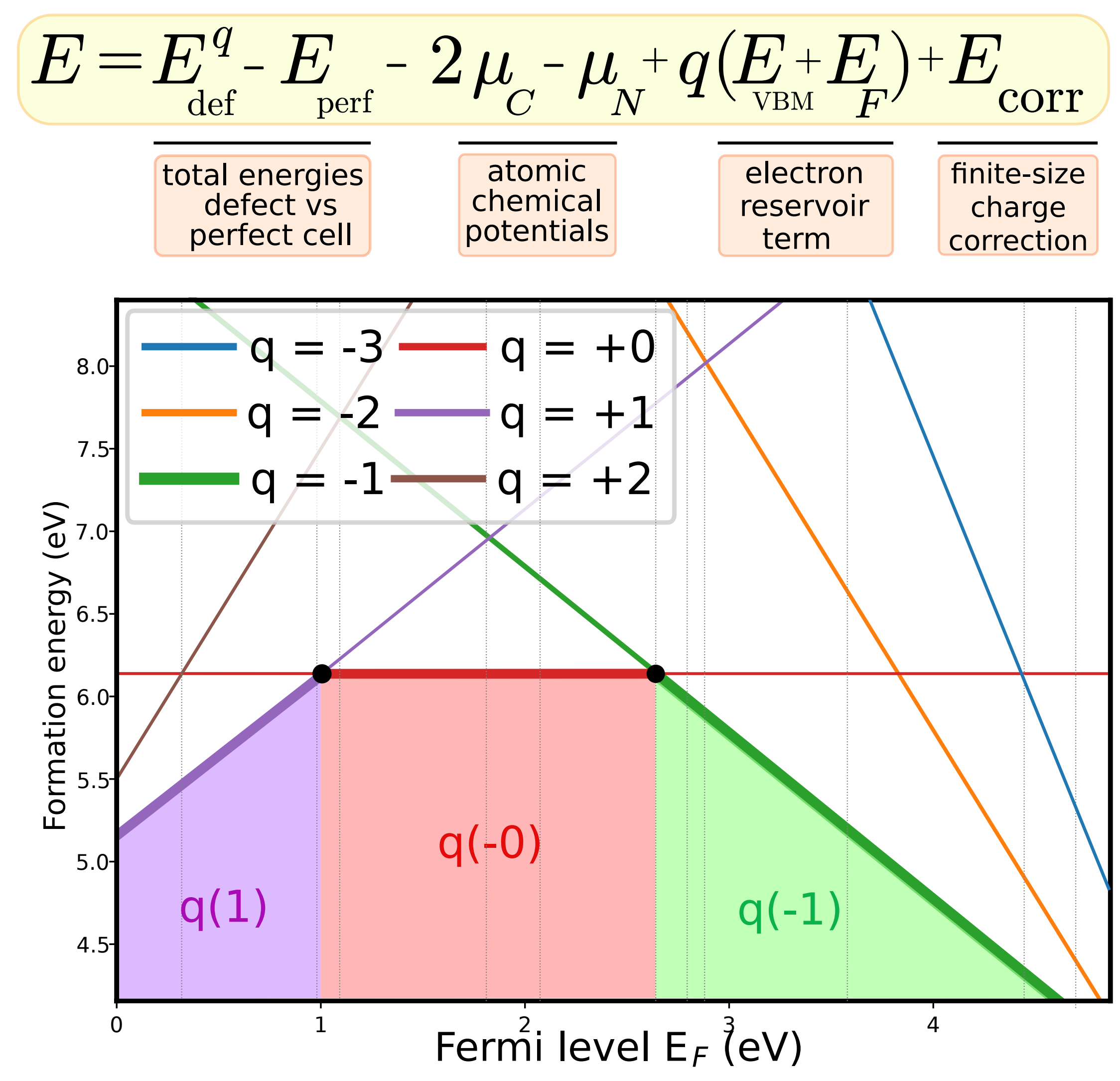
Requirement	Why it matters	Method
Optically active	Can interact with light — enables initialization and readout	LDOS
Highly localized states	Individually addressable — a single, well-defined qubit	Kohn-Sham diagram
Levels inside the band gap	Isolated from bulk bands — accessible, distinguishable transitions	Formation energy
Paramagnetic ground state	Nonzero spin — enables spin-based control and readout	ZPL calculation

Building the Defect, Step by Step



Which Charge State Wins?

To find the thermodynamically favorable charge state, we compute NV's formation energy across all charges ($q = -3$ to $+2$) vs. Fermi level:

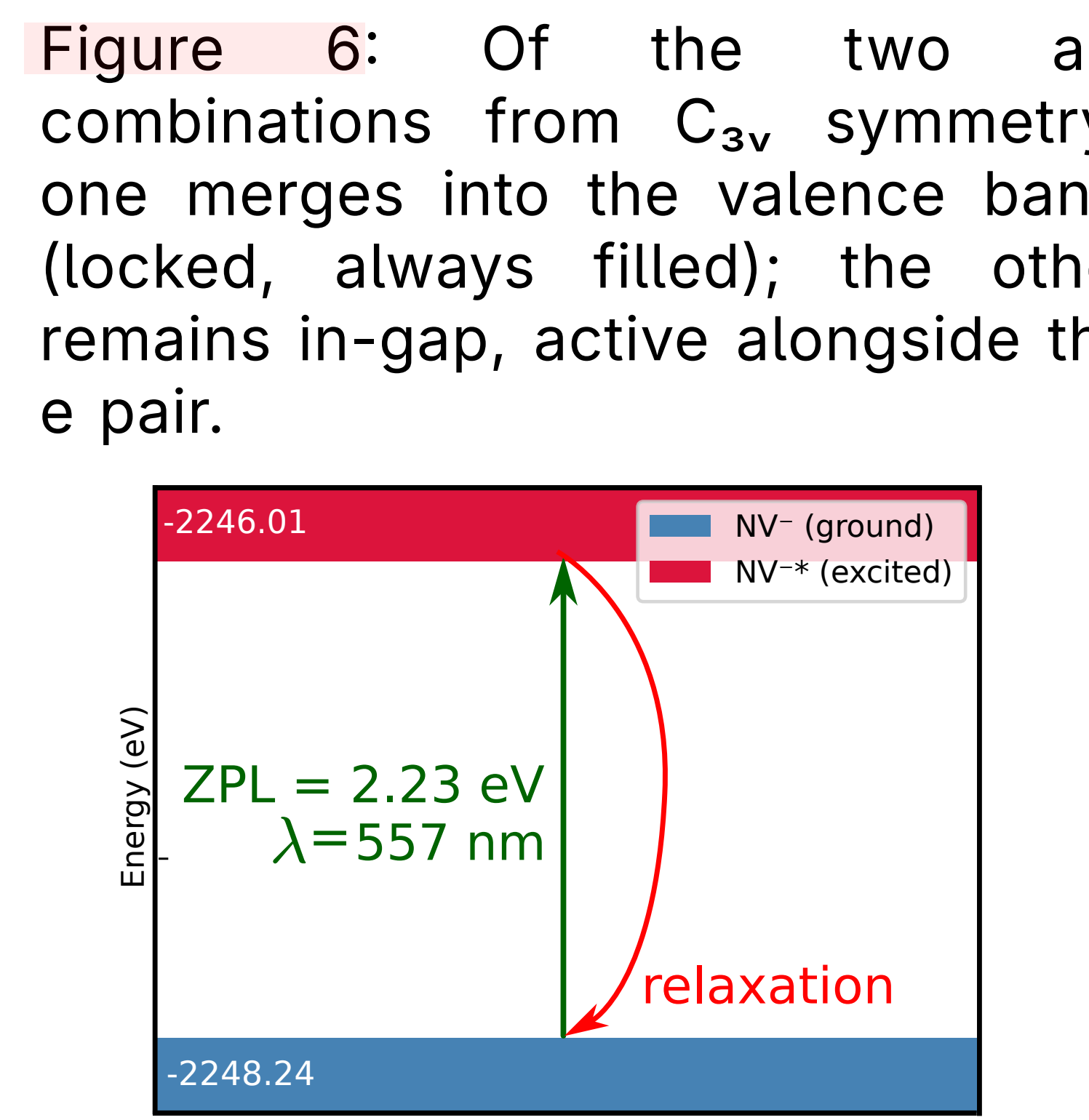
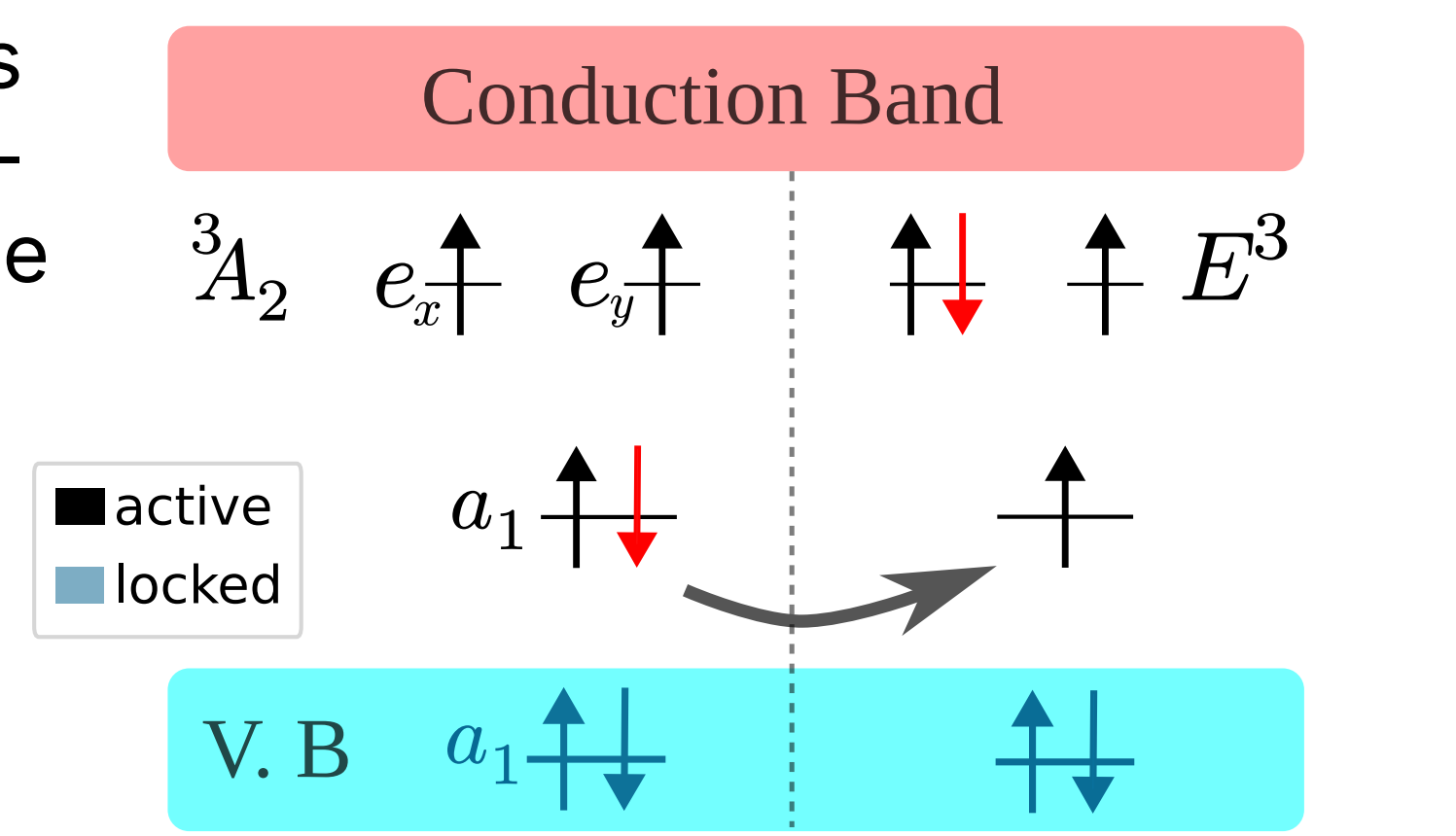
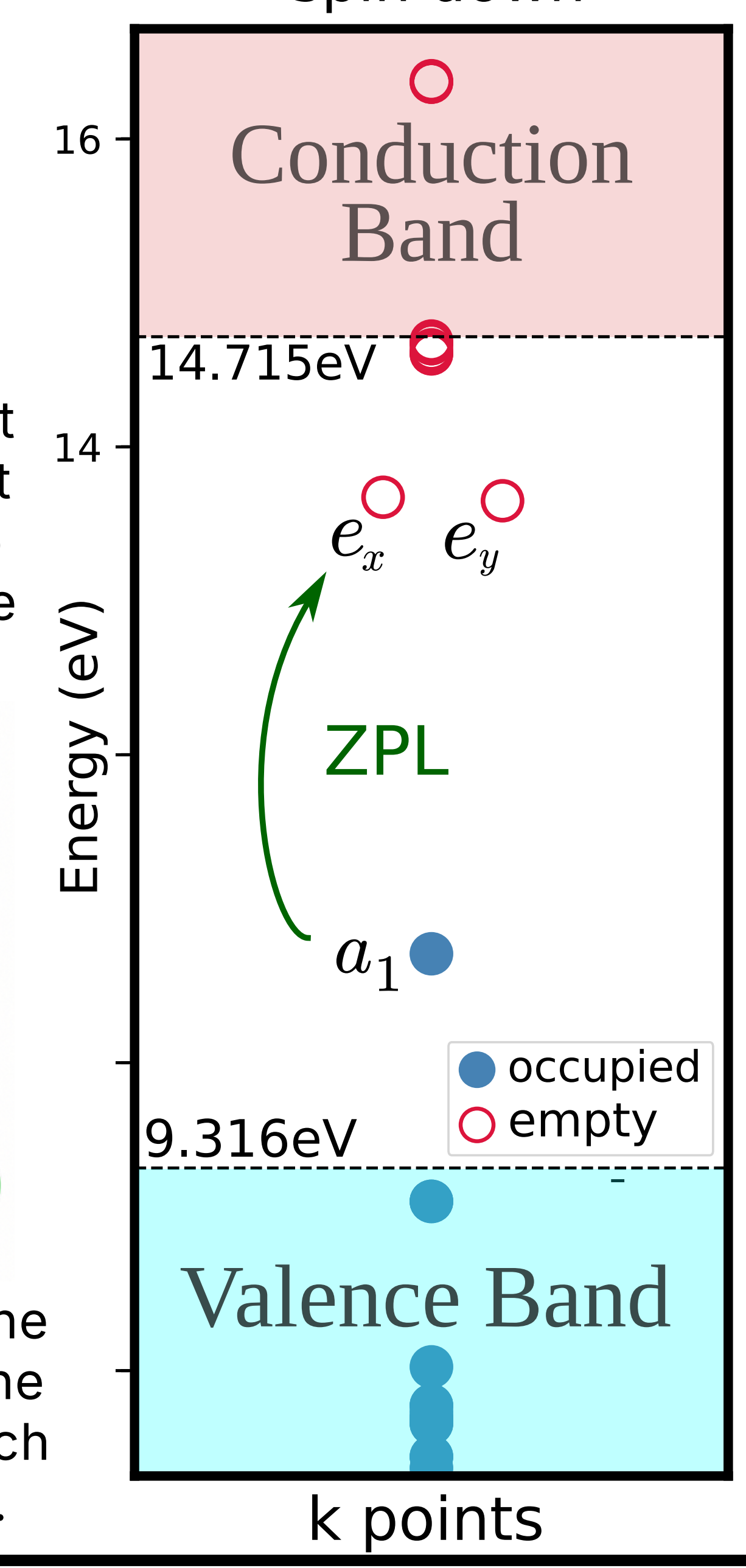
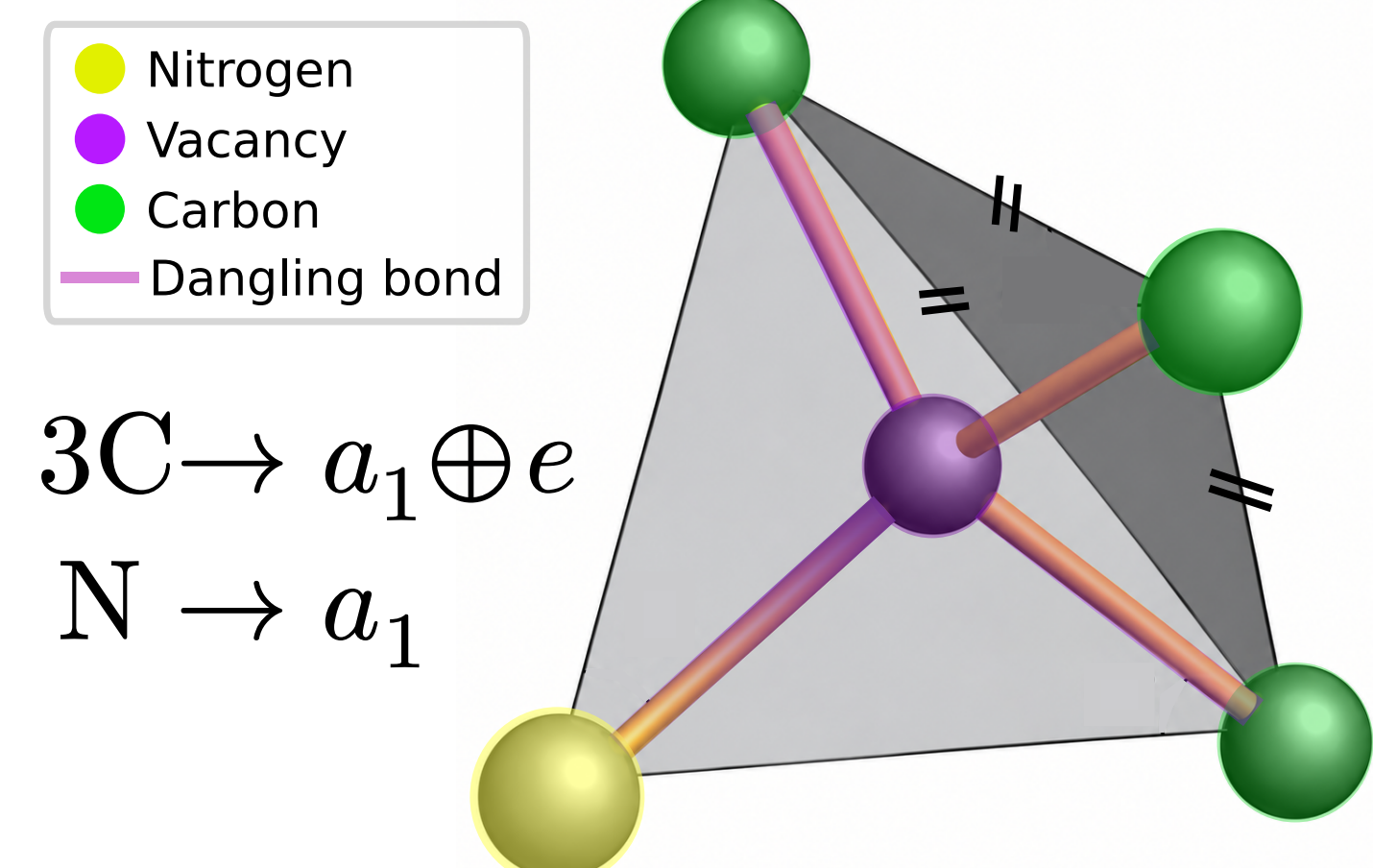


How NV⁻ Talks to Light

Three charge states physically relevant from $q = +1, 0, -1$. Their spin and occupation determine which one is truly qubit-relevant. Figure 5: One electron moves from a_1 (filled) to the doubly-degenerate e level, driving the $^3A_2 \rightarrow ^3E$ optical transition spin down

Charge	Spin (S)	Unpaired e ⁻
+1	0.0	0
0	0.5	1
-1	1.0	2

Table 1: NV⁻ is the only relevant charge state with a spin-triplet ($S=1$) ground state — the two unpaired electrons are what make optical spin-readout possible."



Can We Trust the Method?

Quantity	PBE	HSE06	Exp
Lattice constant (Å)	3.5737	3.5481	3.567
Band gap (eV)	4.12	5.34	5.47
ZPL (eV)		2.23	1.945

Table 2: HSE06 corrects PBE's band-gap underestimation, justifying its use for defect calculations

Quantity	This work	Literature
Formation energy $E_F = 0$ (eV)	8.78	
transition level $\epsilon(0/-)$	2.64	

Table 3: Formation energies depend on the chemical potential reference used and are best compared within the same convention.

What This Means For Quantum Sensing

NV⁻ is both thermodynamically stable and optically addressable — properties already exploited in real-world NV-diamond quantum sensors — and this work provides first-principles confirmation of why NV⁻ is the charge state responsible.

- Microwave control of the ground-state spin sublevels enables coherent qubit gate operations, complementing the optical readout shown here
- Point defects like the NV center are emerging as a strong platform bridging quantum sensing, computation, and scalable room-temperature devices

References

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